Interface Design Basics

Chapter 5 Alax dix

Topics

- Four Golden Rules of Interface Design (Navigation Design)
- Screen Design and Layout

Four Golden rules

- knowing where you are (Current Position)
- knowing what you can do (Available actions at current position)
- knowing where you are going (knowing expected result)
  - or what will happen
- knowing where you’ve been (Past Path)
  - or what you’ve done

These rules are the foundation of good interface design.

If we use them, user will not feel lost, confused, or frustrated.

Why These rules matter:

- Prevent Confusion, Build mental map of the system, improve confidence (rule 1)
- Reduce Hesitation and frustration (Rule 2)
- Reduces uncertainty, Prevents Mistake, and build trust (Rule 3)
- Recover from mistakes/errors (rule 4)

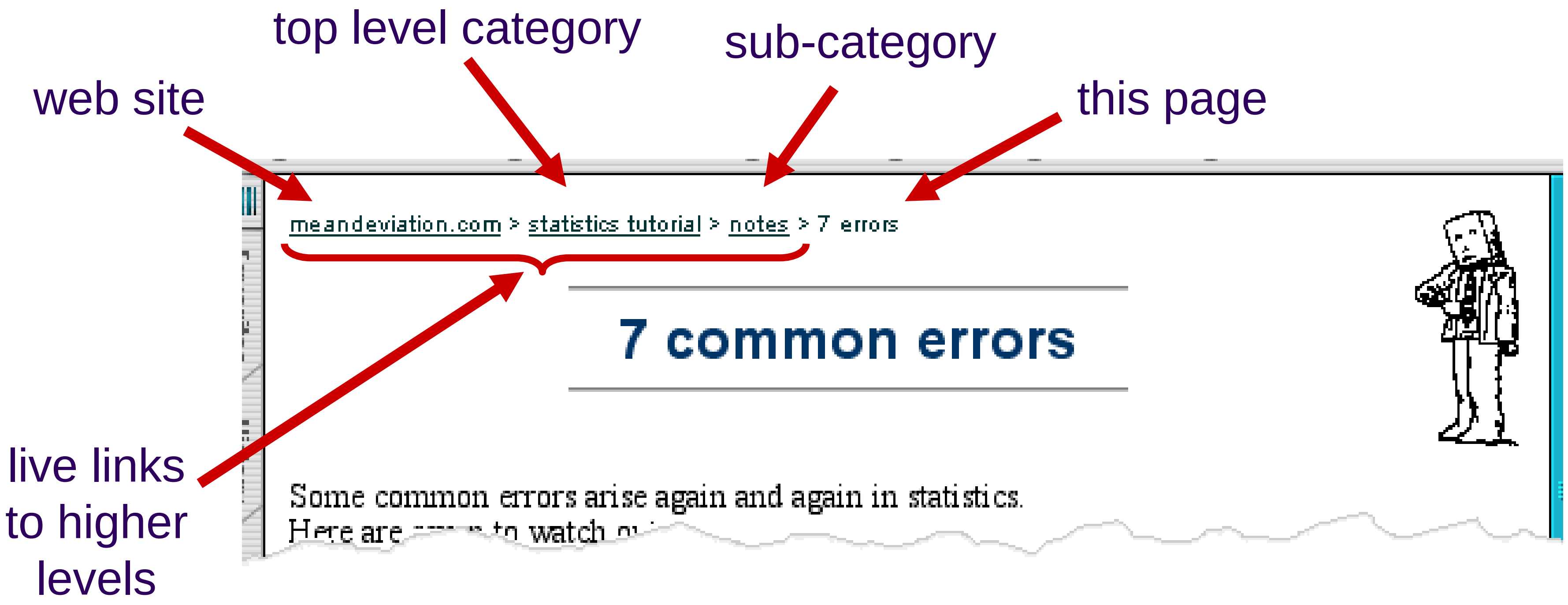
Good Navigation ensures that the user always understand their current position, available actions, likely/expected results, and past path

Knowing **Where You Are** (Current Context)

- Users should always know their **current location within the system**
- interface should clearly indicate the **current page, state, or step** in a process.
- Without this, users may feel lost, especially in complex systems.
  - Page titles
  - Highlighted navigation menu
  - Breadcrumb navigation
  - Step indicators (Step 2 of 5)
  - Section headers

where you are – breadcrumbs

shows path through web site hierarchy

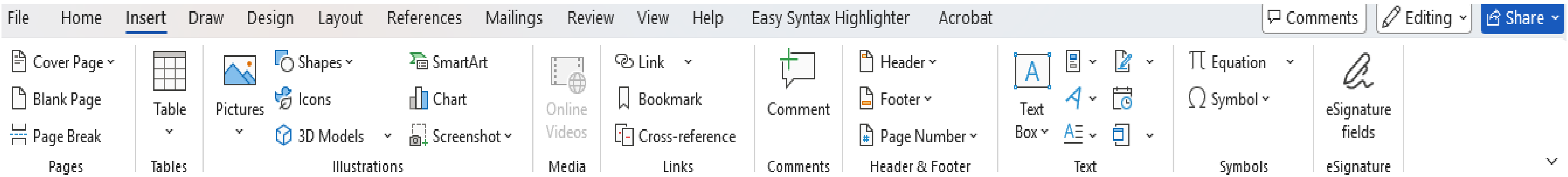


Knowing **What You Can Do** (Available Actions)

- What actions are possible at any point in the interface
- If users cannot see what they can do, they may feel confused or frustrated.
- **Common Design Techniques**
  - Clear buttons and labels
  - Visible menus
  - Tooltips
  - Icons with text labels
  - Disabled buttons for unavailable actions

Knowing **What You Can Do** (Available Actions)

- **Example**
- In a document editor, the toolbar clearly shows actions such as:
- Save , Edit , Undo , Print , Share
- User know what can be inserted



Knowing **Where You Are Going**

- **What Will Happen** (System Feedback and Predictability)
- Users should be able to **predict the outcome of their actions.**
- This reduces uncertainty and improves user confidence.
- **Common Design Techniques**
  - Confirmation dialogs
  - Progress indicators
  - Preview features (WYSIWYG -> What you see is what you get)
  - Informative button labels

Knowing **Where You Are Going**

- **Example**
- When clicking **“Delete File”**, the system shows a confirmation message:
- “Are you sure you want to delete this file? This action cannot be undone.”
- This informs the user what will happen before the action is completed.

Knowing **Where You’ve Been**

- **What You’ve Done** (Interaction History)
- Users should be able to **review past actions or navigation history.**
- helps users recover from mistakes.
- maintain orientation within the system
- **Common Design Techniques**
  - **Browser history**
  - **Undo/Redo (photoshop -> ctrl+z , ctrl+shift+z)**
  - **Activity logs**
  - **Recently viewed items**
  - **Highlighted visited links**

Knowing **Where You’ve Been**

- **Example**
- In a web browser:
- The **Back button**
- **Visited links change color**
- These features help users understand their navigation history.

Class Activity

- Map four golden rules on Online Learning Management System

Knowing Where You Are

- The system shows navigation such as:
- **Dashboard > Course: Software Engineering > Assignment 2**
- Students always know which course and section they are currently viewing.

Knowing What You Can Do

- Each course page shows clear options:
- View lectures
- Submit assignment
- Join discussion forum
- Check grades
- These actions are clearly visible through buttons or menus.

Knowing Where You Are Going

- When a student clicks **Submit Assignment**, the system shows:
- File upload screen
- Confirmation message after submission
- Status: “Assignment Submitted Successfully”
- This informs the student what will happen after the action.

Knowing Where You′ve Been

- The LMS provides:
- List of **previously viewed lectures**
- Assignment submission history
- Activity log
- Students can review what work they have completed

What question we have answered by mapping golden rules of interface design to Online LMS?

-> **what to show**

Next question is how should we present it on the screen

Screen Design and Layout

Dix , Alan
Finlay, Janet
Abowd, Gregory
Beale, Russell

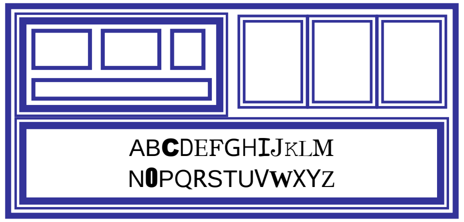
screen design and layout

basic principles

grouping, structure, order

alignment

use of white space



basic principles

- ask
  - what is the user doing?
- think
  - what information, comparisons, order
- design
  - form follows function

available tools

- grouping of items
- order of items
- decoration - fonts, boxes etc.
- alignment of items
- white space between items

grouping and structure

logically together ⇒ physically together

Billing details:
Name
Address: …
Credit card no

Delivery details:
Name
Address: …
Delivery time

Order details:

item	quantity	cost/item	cost
size 10 screws (boxes)	7	3.71	25.97
……	…	…	…

Item belongs together should be placed together

why:

- Helps quick understanding
- Reduce cognitive load

=> If items are logically related, they should be visually related

order of groups and items

- think! - what is natural order

=> arrange the elements in the order user expect

- should match screen order!
  - use boxes, space etc.
  - set up tabbing right!

arrange the elements in the order user expect

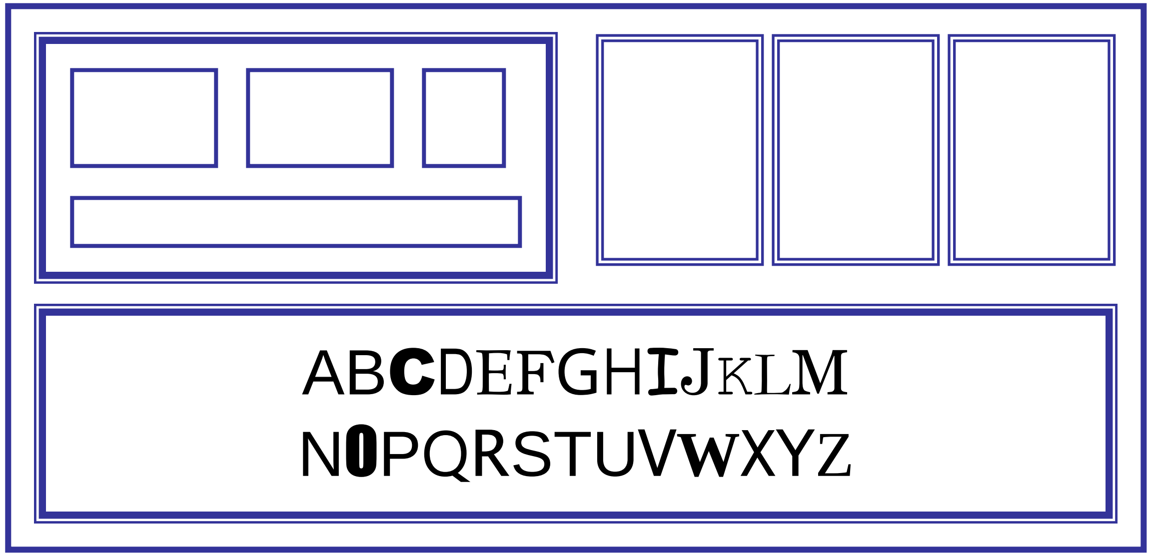
why:

- Matches user thinking
- Reduce confusion

The screen order should match the user’s natural workflow

decoration

- use boxes to group logical items
- use fonts for emphasis, headings, colors
- but not too many!!



Use the visual elements carefully

why: Decoration should support understanding, not distract from it.

alignment - text

- you read from left to right (English and European)

  ⇒ align left hand side

Willy Wonka and the Chocolate Factory
Winston Churchill - A Biography
Wizard of Oz
Xena - Warrior Princess

boring but readable!

Willy Wonka and the Chocolate Factory
Winston Churchill - A Biography
Wizard of Oz
Xena - Warrior Princess

fine for special effects but hard to scan

Align elements properly to improve readability

=> it is not decoration, it improves readability and comparison

alignment - names

- Usually scanning for surnames ⇒ make it easy!

Alan Dix
Janet Finlay
Gregory Abowd
Russell Beale

Alan	Dix
Janet	Finlay
Gregory	Abowd
Russell	Beale

Dix , Alan
Finlay, Janet
Abowd, Gregory
Beale, Russell

alignment - numbers

think purpose!

which is biggest?

532.56
179.3
256.317
15
73.948
1035
3.142
497.6256

alignment - numbers

visually:
long number = big number

align decimal points
or right align integers

627.865
1.005763
382.583
2502.56
432.935
2.0175
652.87
56.34

alignment - numbers

think purpose!

which is biggest?

532.560
179.300
256.317
15.000
73.948
1035.00000
3.142
497.6256

multiple columns

- scanning across gaps hard:

(often hard to avoid with large data base fields)

sherbert   75
x1
toffee 120   x2
chocolate   35
x3
fruit gums   27
x4
coconut dreams   85
x5

multiple columns - 2

- use leaders

sherbert .... 75 .... h
toffee 120 x
chocolate .... 35 ....
fruit gums .... 27 ....
coconut dreams .... 85 ....

multiple columns - 3

- or greying (vertical too)

sherbert	75
toffee	120
chocolate	35
fruit gums	27
coconut dreams	85

multiple columns - 4

- or even (with care!) ‘bad’ alignment

sherbert 75
toffee 120
chocolate 35
fruit gums 27
coconut dreams 85

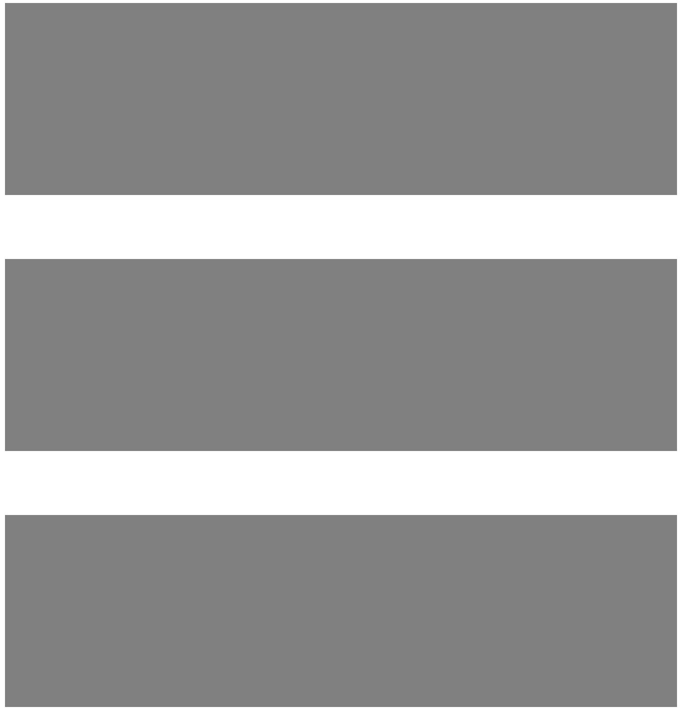
Design should support easy scanning especially in tabular data

White space - the counter

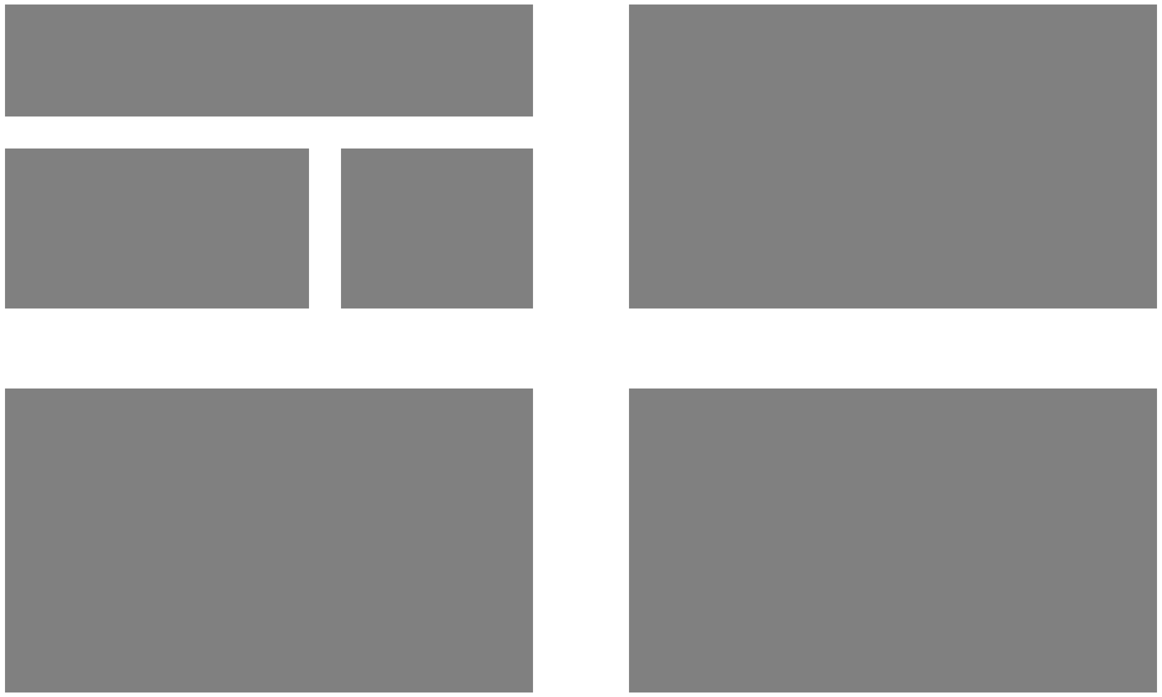
- If related elements look separate, then something is wrong
- Screwing your eyes so that the screen becomes slightly blurred

-Helps separate elements , organize layouts, OR highlight important parts

space to separate



space to structure



space to highlight

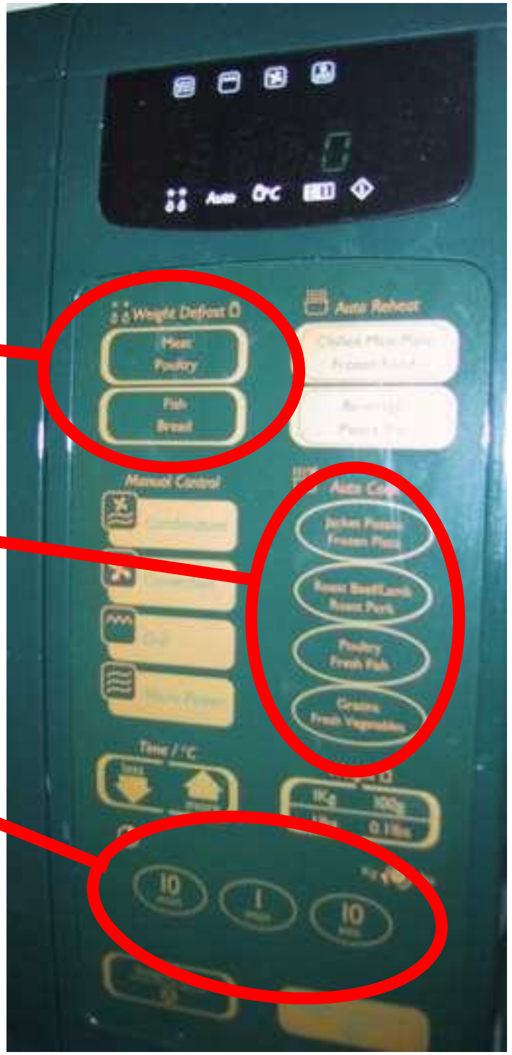
physical controls

- grouping of items
  - defr
  - type
  - time

defrost settings

type of food

time to cook

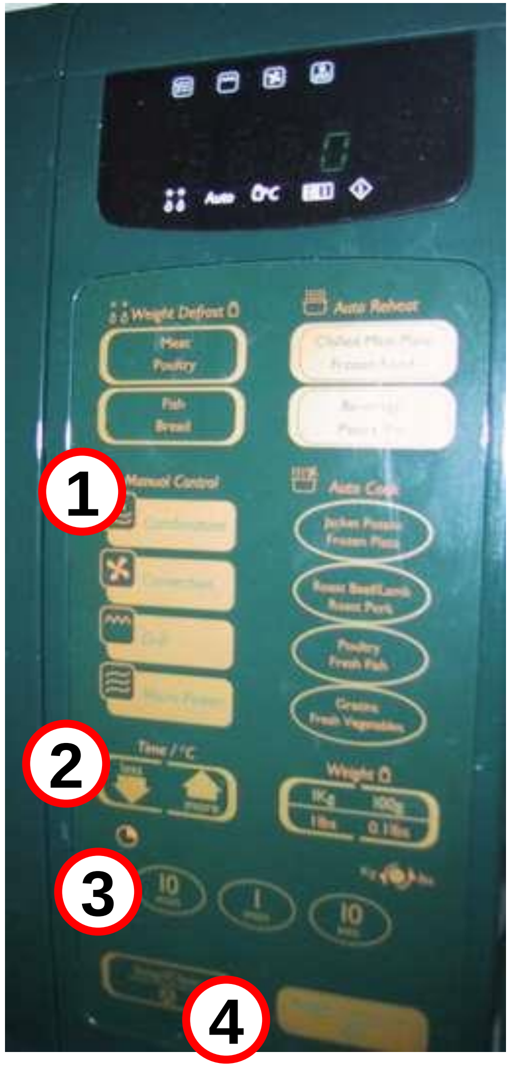


physical controls

- grouping of items
- order of items

1) type
2) tem
3) time
4) start

1) type of heating
2) temperature
3) time to cook
4) start

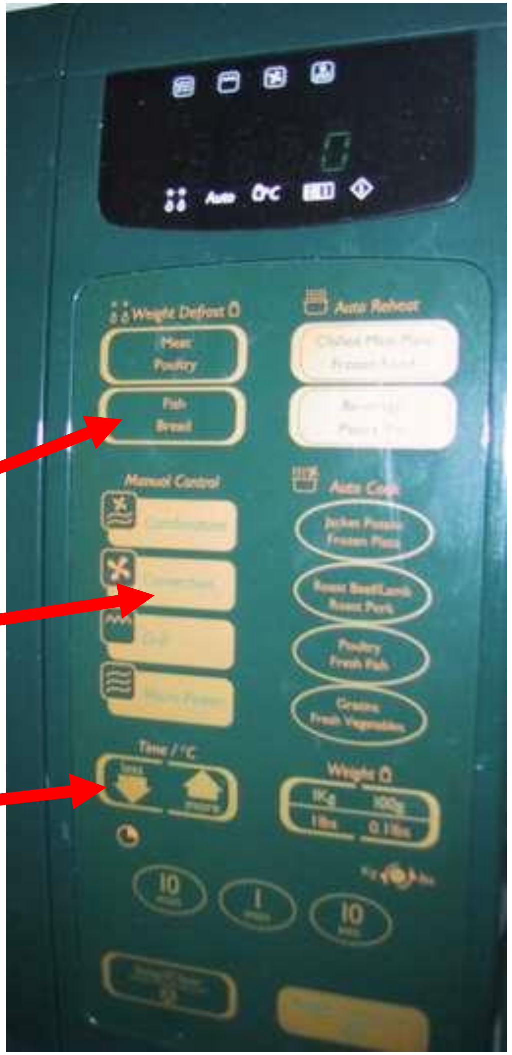


physical controls

- grouping of items
- order of items
- decoration
  - differ
    for di
  - lines
    butto

different colours for
different functions

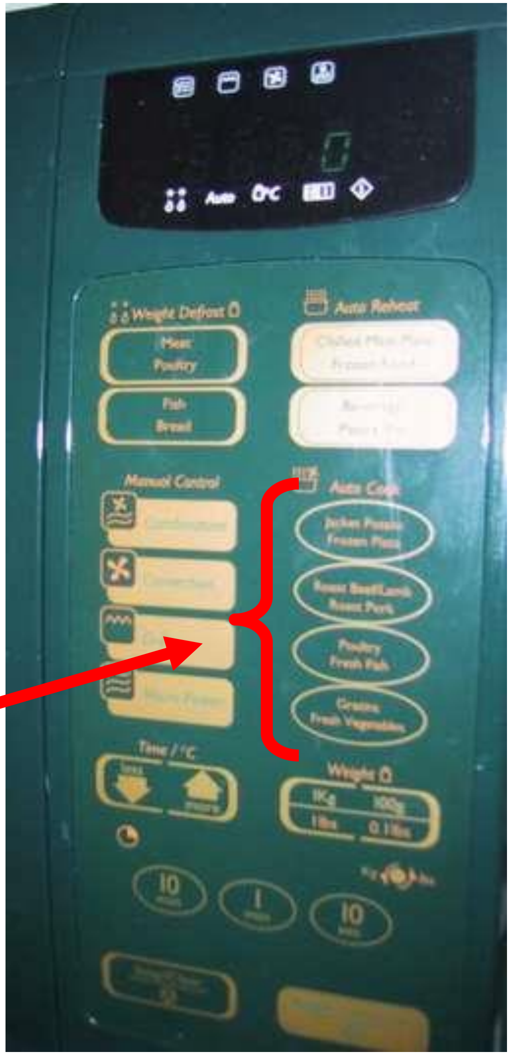
lines around related
buttons (temp up/down)



physical controls

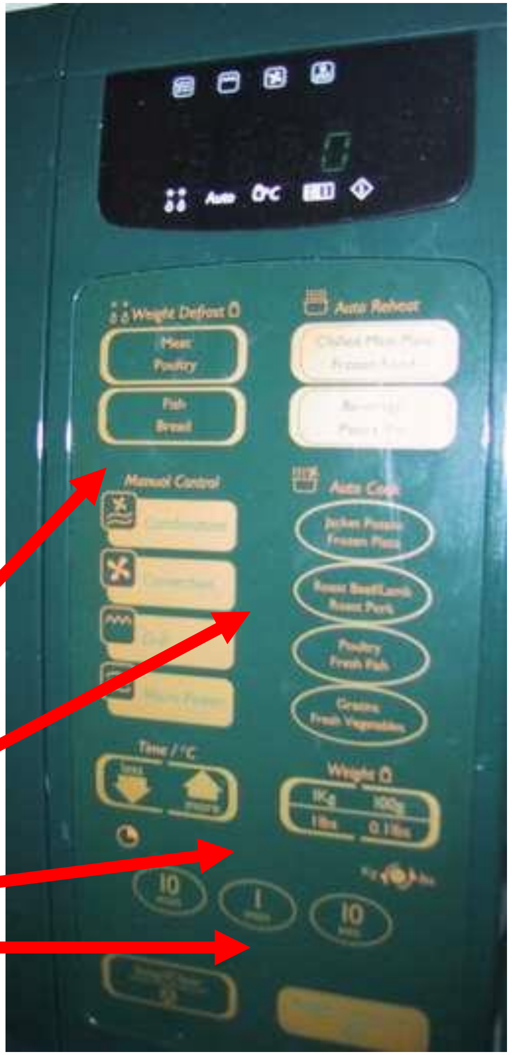
- grouping of items
- order of items
- decoration
- alignment
  - cente
    ? eas

centred text in buttons



physical controls

- grouping of items
- order of items
- decoration
- alignment
- white space
  - gaps t



gaps to aid grouping

Interface design principle apply to both physical and digital systems

Good interface design ensures user always understands where they are, what they can do, what will happen, and what they have already done.

these principles are supported by proper screen layout techniques (grouping, order, alignment, white space, decoration) => making the system easy, efficient to use